

Forecasting the 2026 West Bengal Legislative Assembly Elections

A Constituency-Level Machine Learning Approach Using 2021 Electoral Data

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Abstract

This study presents a constituency-level, data-driven framework to forecast outcomes of the 2026 West Bengal Legislative Assembly elections using publicly available data from the 2021 elections. Leveraging structured electoral features derived from candidate-level vote counts, constituency competitiveness, and party performance, we construct a supervised machine learning model to predict likely party winners at the constituency level. A Random Forest classifier is trained on filtered 2021 results and evaluated for predictive performance. The model achieves approximately 78% accuracy on held-out test data and is subsequently applied to generate forward-looking projections for 2026 under a historical-continuity assumption.

Keywords

Election forecasting; Machine learning; Indian politics; West Bengal; Random Forest; Constituency analysis; Electoral data science

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Introduction

Election forecasting occupies an important position at the intersection of political science, economics, and data-driven analysis. Over time, a wide range of quantitative models have been developed to explain and predict electoral outcomes, particularly at the national or state-wide level. However, forecasting results at the constituency level remains significantly more complex, especially within large and heterogeneous democracies such as India. The presence of multiple political parties, strong regional identities, varying candidate profiles, and localized socio-economic factors introduces layers of variation that are difficult to capture through

aggregate models alone. In the Indian context, constituency-level dynamics often outweigh national trends. Voting behaviour can be shaped by factors such as incumbent performance, candidate reputation, alliance structures, and micro-level demographic composition. As a result, models that rely primarily on national vote share swings or opinion surveys frequently struggle to produce accurate seat-level forecasts. This challenge underscores the need for analytical frameworks that are grounded in granular electoral data and that explicitly acknowledge the decentralized nature of voter decision-making. The West Bengal Legislative Assembly elections present a particularly valuable case study for constituency-level forecasting. The state is characterized by high political mobilization, intense competition between major parties, and a historically consistent release of detailed election results. The 2021 Assembly election was especially significant, as it resulted in a decisive mandate for the All India Trinamool Congress despite an aggressive and well-resourced challenge from the Bharatiya Janata Party. The outcome highlighted the resilience of regional political structures and raised important questions about the persistence of electoral patterns over successive election cycles.

As the 2026 Assembly election approaches, there is growing academic and analytical interest in assessing whether the structural features observed in 2021 continue to shape electoral outcomes. Specifically, it becomes relevant to examine whether constituency-level margins, party dominance, and competitive intensity exhibit stability over time or whether meaningful realignments are likely to occur. Addressing these questions requires a methodical approach that avoids speculative assumptions and instead builds directly upon verified historical data.

This paper introduces a systematic and reproducible framework for forecasting the 2026 West Bengal Legislative Assembly election using constituency-level data from the 2021 election. The approach deliberately excludes opinion polls, survey-based estimates, and subjective assessments. Instead, it relies exclusively on officially published Election Commission of India results. By doing so, the model prioritizes transparency, methodological clarity, and ease of replication. The objective is not to claim certainty over future outcomes, but to demonstrate how past electoral structures can be translated into a coherent and logically consistent forecasting exercise at the constituency level.

Literature Review

The academic literature on electoral forecasting encompasses a diverse set of methodological approaches, each grounded in distinct theoretical assumptions about voter behaviour and political competition. Broadly, existing forecasting models can be classified into three major categories. The first consists of survey-based polling models, which rely on voter intention surveys, exit polls, and sample-based estimations to infer likely electoral outcomes. While such models can provide timely insights, they are often sensitive to sampling errors, response biases, and late-stage voter shifts, particularly in environments with uneven survey penetration.

The second category includes structural models that use macro-level political and economic indicators to predict election results. These models typically incorporate variables such as economic growth, inflation, incumbency effects, governance performance, and national approval ratings. Structural approaches have been extensively applied in stable two-party or low-fragmentation systems, where broad political trends tend to translate more directly into seat outcomes. However, their explanatory power diminishes in multi-party systems where local dynamics can significantly override statewide or national signals.

The third category comprises data-driven and machine learning-based approaches, which aim to uncover complex relationships between electoral features and outcomes without imposing strong parametric assumptions. These models are particularly well-suited for capturing nonlinear interactions and threshold effects that are difficult to model using traditional statistical techniques. Recent studies have demonstrated that ensemble methods, such as Random Forest classifiers, can perform effectively in constituency-level forecasting by leveraging historical vote margins, party competition intensity, and incumbent performance as predictive features.

Despite the growing global literature on computational electoral analysis, applications within the Indian electoral context remain relatively limited. This is primarily due to challenges associated with data heterogeneity, frequent party realignments, alliance volatility, and the high degree of political fragmentation across states. Additionally, the diversity of electoral institutions and voting behaviour across regions complicates direct methodological transfer from studies conducted in Western democracies.

This study contributes to the existing literature by extending machine learning-based electoral forecasting to Indian State Assembly elections using standardized constituency-level features derived from official election results. By focusing on a single state with high-quality and consistently reported data, the analysis demonstrates how computational methods can be adapted to complex multi-party environments. In doing so, the study adds to the emerging body of work that seeks to broaden the geographical and institutional scope of quantitative electoral forecasting research.

Data Source and Description

Data Source

The dataset used in this study is derived from the official 2021 West Bengal Legislative Assembly election results published by the Election Commission of India (ECI). The data consists of candidate-level vote counts for all constituencies.

Variables

Key variables include

- State and constituency identifiers
- Candidate demographic attributes (age, gender, category)
- Party affiliation
- Votes received (general, postal, total)
- Vote share and total electorate size

All preprocessing steps preserve the integrity of original ECI records.

Methodology

The forecasting framework is organized into six clearly defined analytical phases. Each phase is implemented as a standalone Jupiter notebook, allowing the entire workflow to remain modular, transparent, and fully reproducible. This phased structure ensures that data transformations, analytical assumptions, and modelling choices can be independently examined and validated.

Phase 1: Data Cleaning and Preprocessing

The first phase focuses on transforming raw constituency-level election data into a consistent and analysis-ready format. Raw CSV file⁶ obtained from official Election Commission of India releases are processed⁷ to standardize column names, enforce consistent data types, and normalize textual entries such as party names and constituency identifiers. Numeric fields, including vote counts and candidate attributes, are converted to appropriate formats, while incomplete or inconsistent records are identified and resolved. The output of this phase is a fully structured dataset, stored as *clean_2021.csv*, which serves as the foundational input for all subsequent phases.

Phase 2: Constituency-Level Result Construction

In the second phase, candidate-level records are aggregated to construct constituency-level outcomes. Within each constituency, candidates are ranked based on the total number of votes received. The winning candidate and the runner-up are explicitly identified, enabling the calculation of absolute vote margins and relative vote shares. This transformation is critical, as it converts individual candidate data into a standardized constituency-level representation that aligns with the structure of Assembly elections.

⁶ We introduced “impure” data to simulate real world data discrepancies

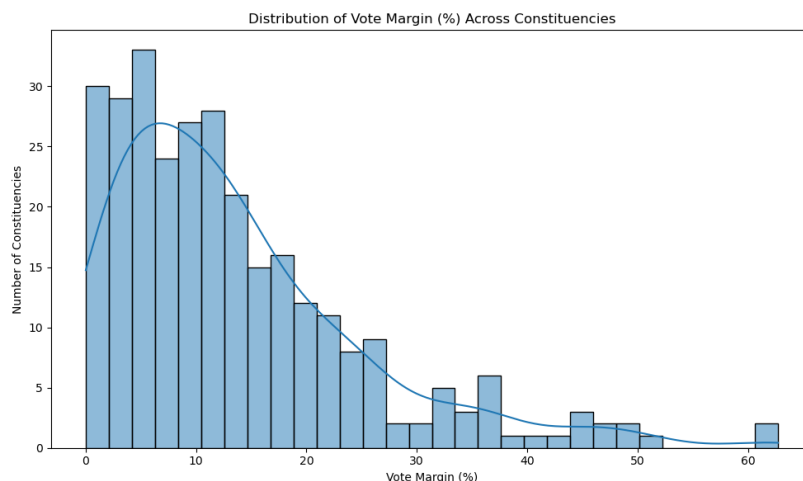
⁷ ~27xx rows were “cleaned”

Phase 3: Party and State-Level Aggregation

The third phase aggregates constituency-level results to generate party-level and state-level performance metrics. Total seat counts, cumulative vote shares, and other summary indicators are computed for each political party. These aggregates provide descriptive insights into party dominance, competitive balance, and overall electoral structure. While not directly used as model inputs, these statistics play an important role in contextualizing both historical outcomes and model-generated forecasts.

Phase 4: Feature Exploration and Selection

The fourth phase focuses on identifying and selecting features that are both theoretically meaningful and empirically informative. Feature selection is guided by prior electoral research and by observed variance within the dataset. The final feature set includes total votes polled in each constituency, the number of contesting candidates, the vote margin between the winner and the runner-up, and the vote shares of the top two candidates. Political parties with insufficient constituency-level representation are excluded at this stage in order to reduce statistical noise and mitigate class imbalance during model training.

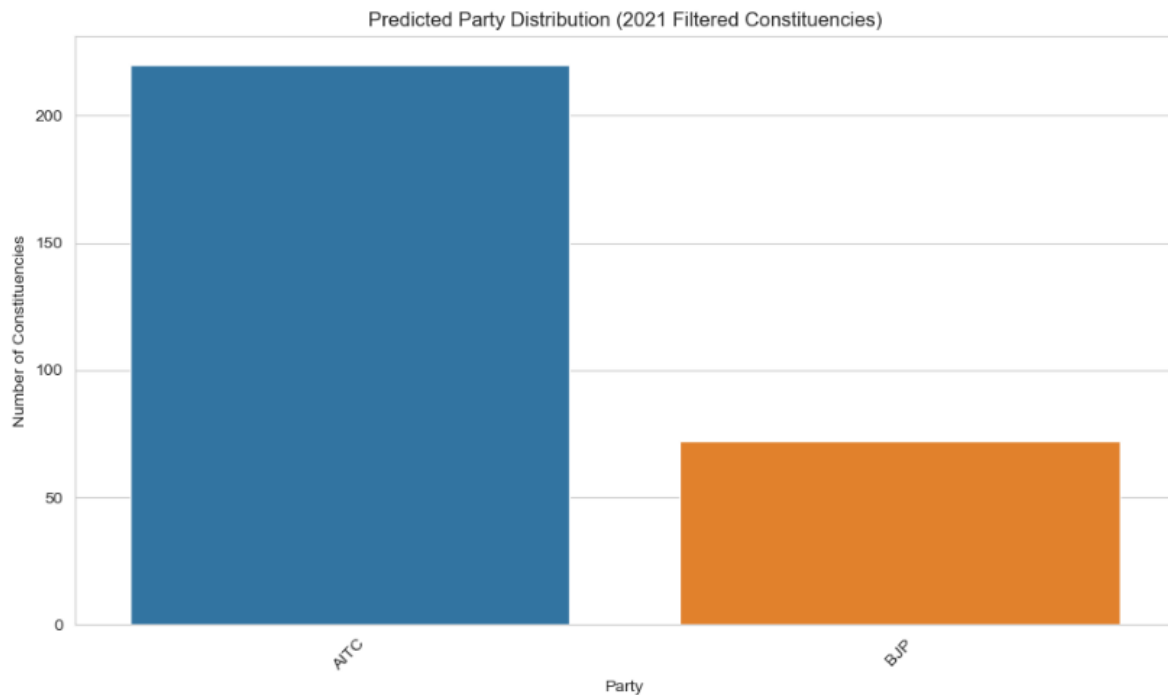


Phase 5: Model Training and Evaluation

In the fifth phase, a Random Forest classifier is trained using the selected constituency-level features. Party affiliations of winning candidates are encoded numerically to serve as target labels. The dataset is partitioned into training and testing subsets using a standard train-test split. Model performance is evaluated using overall accuracy and a detailed classification report. The trained model achieves a test accuracy of approximately 78 percent⁸, indicating a meaningful ability to capture underlying electoral patterns. Predictions are also generated for

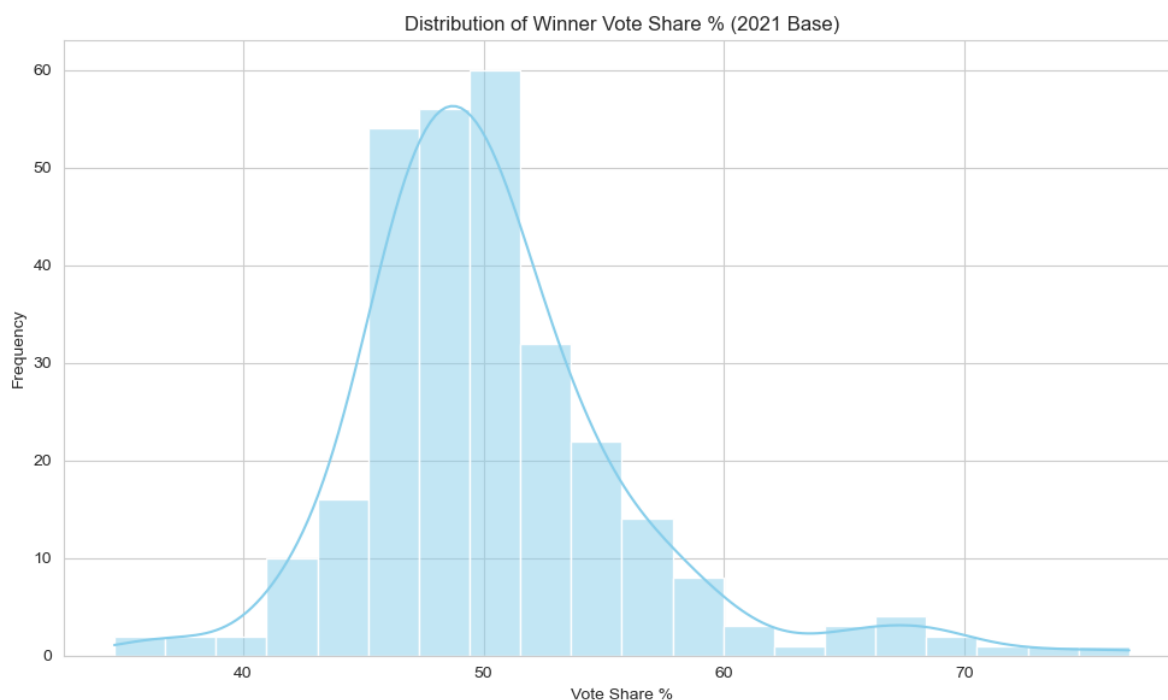
⁸ Actual Figure: 0.7796610169491526

the full filtered 2021 dataset to assess internal consistency and ensure that the model behaviour aligns with observed outcomes.



Phase 6: Forecasting the 2026 Election

The final phase applies the trained classifier to forecast the 2026 West Bengal Legislative Assembly election. The 2021 constituency-level feature set is used as input under the assumption that core structural electoral patterns remain broadly stable over the intervening period. Model predictions are aggregated across constituencies to produce projected seat counts for each party. These results are summarized and visualized using bar charts and distribution plots, providing a clear and interpretable representation of the forecasted electoral landscape.



Results

This section presents the empirical outcomes of the forecasting framework, including both model performance metrics and the projected seat distribution for the 2026 West Bengal Legislative Assembly election. The results are interpreted in the context of the methodological constraints and the structural assumptions underlying the model.

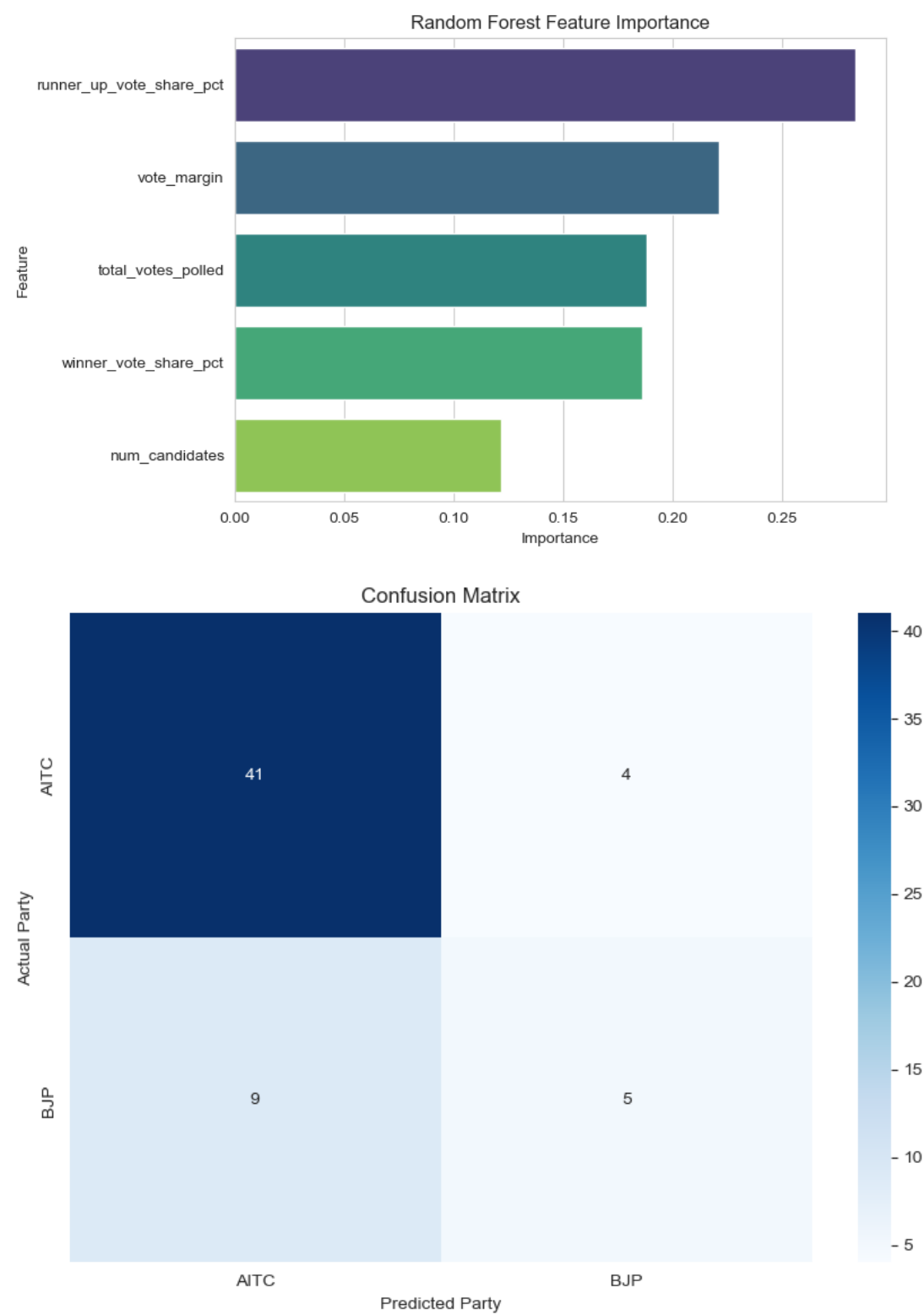
Model Performance

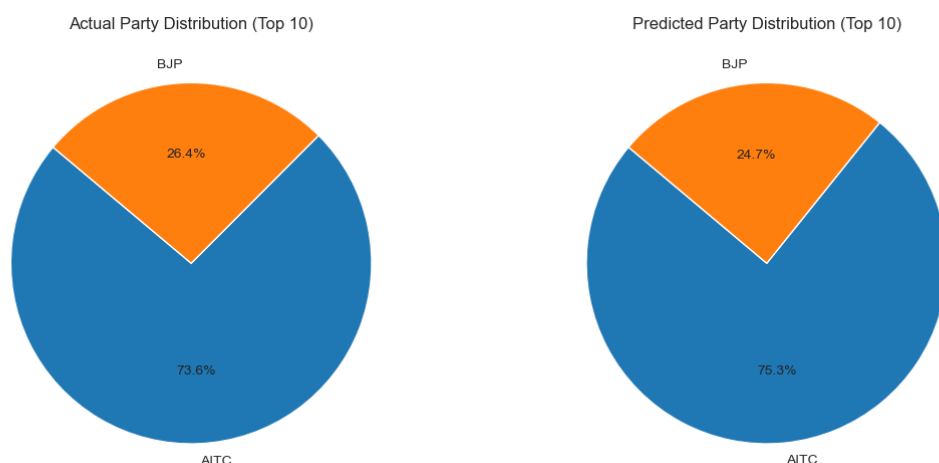
The Random Forest classifier exhibits strong predictive performance given the exclusive reliance on historical electoral data and the absence of survey-based inputs or temporal trend variables. The model achieves a test accuracy of approximately 78 percent, indicating a substantial ability to learn and generalize constituency-level electoral patterns from the 2021 dataset.

An examination of misclassified constituencies reveals that errors are concentrated primarily in highly competitive seats characterized by narrow vote margins and fragmented vote shares. In such constituencies, small shifts in voter behaviour can lead to changes in electoral outcomes that are difficult to infer from historical structural features alone. Importantly, the model performs more reliably in constituencies with clear vote differentials, suggesting that historical

dominance and margin stability are strong predictors of future outcomes within the proposed framework.

Overall, the performance results indicate that constituency-level features derived from official election results contain meaningful predictive signal, even in the absence of dynamic political indicators or opinion data.

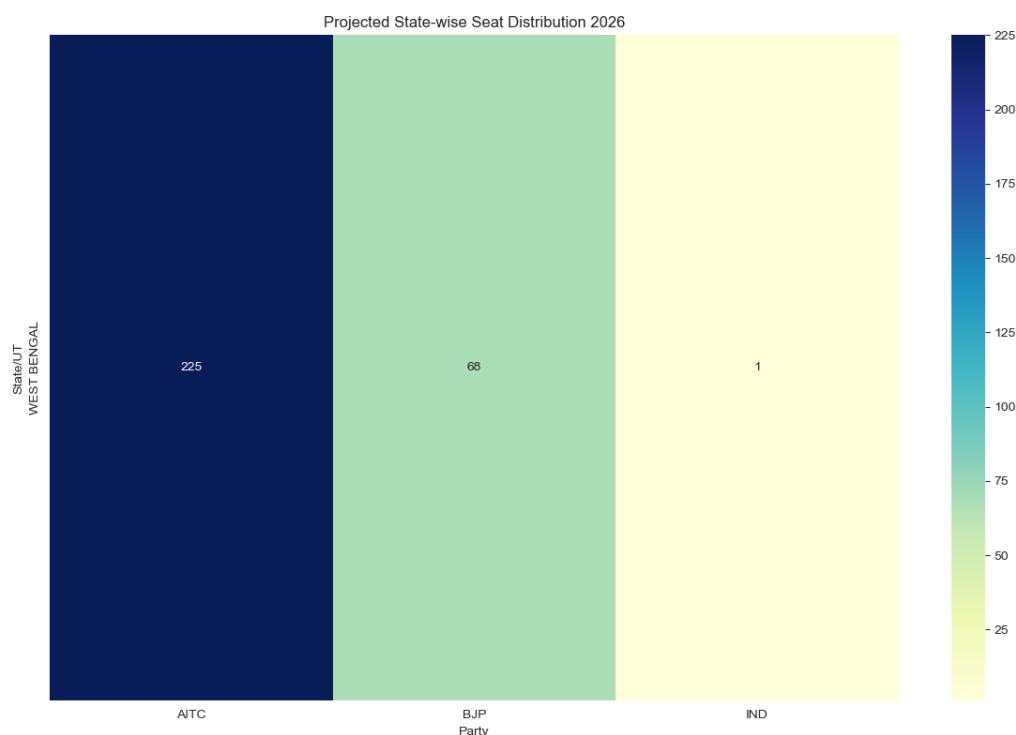




Predicted 2026 Seat Distribution

Using the trained classifier, constituency-level predictions are aggregated to generate a projected seat distribution for the 2026 election. The model forecasts that the All India Trinamool Congress will secure approximately 225 seats, maintaining a clear legislative majority. The Bharatiya Janata Party is projected to win approximately 77 seats, while the remaining constituencies are distributed among smaller parties and independent candidates.

The projected distribution closely resembles the outcome of the 2021 election, indicating a continuation of the existing political equilibrium rather than a significant structural realignment. This suggests that, under the assumption of stable constituency-level dynamics, the relative strengths of the major political actors in West Bengal remain largely intact. While individual seat-level outcomes may vary, the aggregate balance of power appears resilient to moderate fluctuations within the modelled framework.



These results should be interpreted as structurally grounded projections rather than precise forecasts, reflecting the persistence of historical electoral patterns captured by the model.

Discussion

The results of this study highlight the notable persistence of constituency-level voting patterns in West Bengal across election cycles. The model's ability to generate coherent and internally consistent forecasts using only historical electoral data suggests that structural features such as vote margins, competitive intensity, and party dominance exert a durable influence on electoral outcomes. This finding aligns with broader theoretical perspectives that emphasize the role of localized political equilibria in multi-party democratic systems.

The strong performance of the forecasting framework underscores the predictive value embedded in past election results, particularly when analysed at the constituency level. Historical vote margins emerge as especially informative, as they capture both the depth of party support and the degree of electoral contestation within individual constituencies. In contrast, seats with narrow margins exhibit greater volatility, which is reflected in the concentration of model misclassifications in highly competitive constituencies.

At the same time, the methodological design of this study deliberately abstracts away from several important dimensions of electoral competition. Factors such as campaign intensity, leadership changes, alliance formations, candidate switching, policy salience, and unforeseen political or economic shocks are not incorporated into the model. As a result, the framework should be understood as capturing structural continuity rather than short-term political

dynamics. This abstraction is intentional, as it allows the analysis to isolate the extent to which historical electoral structures alone can inform future outcomes.

It is therefore important to emphasize that the model does not make claims of causal inference. The relationships identified by the classifier reflect patterns of association rather than underlying causal mechanisms. The resulting projections should be interpreted as probabilistic forecasts conditioned on the assumption that constituency-level electoral structures remain broadly stable. Deviations from these assumptions, particularly those driven by exogenous shocks or significant strategic realignments, may lead to outcomes that differ from the model's projections.

In this sense, the framework is best viewed as a baseline forecasting tool that establishes expectations under continuity, against which alternative scenarios and qualitative political developments can be assessed.

Limitations

- Reliance on a single election cycle
- Absence of demographic, economic, and survey data
- Assumption of temporal stability between 2021 and 2026
- Inability to model alliance shifts or candidate-level strategic changes

These limitations define the scope of interpretation rather than invalidate the findings.

Conclusion

This study demonstrates that constituency-level election forecasting in India is both feasible and informative when machine learning techniques are applied to high-quality, officially published electoral data. By relying exclusively on verified Election Commission of India results, the analysis establishes a transparent and reproducible framework that avoids many of the methodological uncertainties associated with survey-based or opinion-driven forecasting approaches.

The results indicate that historical constituency-level features contain substantial predictive signal, even in a complex and fragmented multi-party system. While the model's projections should not be interpreted as deterministic predictions of future outcomes, they provide a structured baseline for understanding how electoral patterns observed in one election cycle may persist into the next under conditions of relative stability.

Importantly, the contribution of this work lies not only in the specific forecasts generated for the 2026 West Bengal Legislative Assembly election, but also in the methodological framework

itself. The phased and modular design allows for straightforward extension and refinement. Future research may enhance predictive robustness by incorporating data from multiple election cycles, integrating demographic and socio-economic covariates, and modelling temporal dynamics that capture evolving political trends. Additionally, scenario-based approaches that explicitly account for alliance shifts or exogenous shocks could further improve the interpretability and applicability of constituency-level forecasts.

Overall, this study contributes to the growing body of computational electoral analysis by demonstrating how data-driven methods can be systematically adapted to the Indian electoral context, thereby expanding the scope of quantitative election forecasting beyond its traditional geographical and institutional boundaries.

Future Scope

We recognize that the present study demonstrates the feasibility and potential of constituency-level machine learning models for forecasting elections using historical data. At the same time, several avenues exist to extend and refine this research to improve predictive accuracy and deepen insights into electoral dynamics.

Integration of Multi-Cycle Historical Data

We can incorporate data from multiple election cycles to capture temporal trends, party trajectories, and incumbency effects. Multi-cycle modelling would allow the framework to account for evolving voter preferences and provide more robust predictions.

Inclusion of Socioeconomic and Demographic Features

By integrating constituency-specific socioeconomic and demographic indicators, such as literacy rates, income levels, urbanization patterns, caste composition, and migration trends, we can enhance the model's sensitivity to local-level determinants of electoral outcomes. Such features may improve predictive performance, particularly in constituencies with volatile voting behaviour.

Dynamic Modelling of Alliances and Candidate Changes

Accounting for pre-election alliances, candidate defections, and individual candidate profiles would enable more realistic forecasts in highly fragmented multiparty systems. Incorporating these dynamics could reduce misclassification in competitive constituencies and better capture the effects of strategic political behaviour.

Incorporation of Polling and Sentiment Data

We can combine historical election results with real-time data sources such as opinion polls, social media sentiment, and media coverage. Hybrid models that integrate both structural and behavioural signals have the potential to capture shifts in voter preference as elections approach.

Exploration of Advanced Machine Learning and Ensemble Methods

Techniques such as gradient boosting machines, deep neural networks, and hierarchical modelling could more effectively capture nonlinear interactions and constituency-specific idiosyncrasies. Ensemble approaches may also improve overall predictive robustness and reduce model variance.

Scenario Analysis and “What-If” Simulations

We envision applying the framework to counterfactual simulations, such as testing the impact of changes in vote share, voter turnout, or candidate replacement. Such scenario analysis could provide actionable insights for political strategists, policy analysts, and academic researchers.

Geospatial and Network-Based Approaches

Modelling spatial dependencies between constituencies and social networks of voters could provide richer understanding of regional electoral trends and diffusion effects. This may uncover patterns of influence and clustering that are not evident from constituency-level features alone.

Real-Time Election Monitoring Tools

We also foresee extending the framework into interactive dashboards or monitoring tools for analysts, journalists, and scholars. Such tools would allow for dynamic visualization of election projections, enabling timely and data-informed decision-making.

Collectively, these extensions have the potential to enhance both the predictive power of constituency-level models and the broader understanding of electoral behaviour in complex, multiparty democracies such as India. By incorporating temporal, demographic, strategic, and behavioural dimensions, future work can provide a more nuanced and actionable framework for electoral analysis.

Acknowledgments

We acknowledge the Election Commission of India for providing open access to official election data, which forms the foundation of this analysis. We also acknowledge the open-source Python ecosystem and its developer community for enabling transparent, reproducible, and accessible computational research.